

CHAPTER 6

Notion of Localization Processes

6.1 Localization in Physical Theories

The operational foundations of physics as a natural science are based on observation and measurement. The abstraction of the measurement process gives rise to the notion of *coordinatization* or *arithmetization*. In this sense the formation of algebraic structures of ‘number-like’ quantities of any particular operational form can be thought of as solutions to corresponding physical measurement problems. In the interpretation of physical theories, our empirical access to the world is coordinatized through the concept of observables. *Observables* denote physical quantities that, in principle, can be measured in the context of appropriate experimental arrangements. The crucial further assumption is that quantities admissible as measured results must be *real numbers*, since it is accepted that the resort to real numbers has the advantage of making our empirical access to the world secure. More concretely, the model adopted to represent the measured values of observables is the real line $\mathbb{R}$ and its powers, specified as a set-theoretic structure of points that are independent and possess the property of infinite distinguishability with absolute precision. In this way, the semantics of the *physical continuum* in the standard interpretation of physical systems theories is associated with the codomain of valuation of observables, that is, the set-theoretic real line model $\mathbb{R}$ and its powers. Due to this assumption, in any experiment performed by an observer, the propositions that can be made concerning an observable are of the type which asserts that the value of an observable lies in some *measurable set* of the real numbers. A proposition of this kind corresponds to an *event* as it is apprehended by an observer using his measuring device.

The stipulation of the set-theoretic real line model $\mathbb{R}$ as the semantic information carrier of the notion of physical ‘continuum’ has posed enormous interpretational problems for quantum theory, and in particular quantum measurement. The root of this problem is situated on the different notions of *localization* characterizing *classical* and *quantum* observables. Overlooking this subtle difference,

referring to localization properties of observables in different physical regimes, is a source of paradoxes and misinterpretations.

First, it is important to clarify the meaning of the notion of localization. Physical observation and measurement presupposes, at the fundamental level, the existence of a *localization process* via which we may extract information related to the local behavior of a natural system. Only on the basis of a localization process does it become possible to discern events and assign an individuality to them. Generally, a localization process is co-implied by the preparation of suitable local reference contexts for measurement. These contexts identify concretely the kind of reference frames used for observation of events. The methodology of observation is effectuated by the functioning of event-registering measurement devices, which operate locally within the context of prepared reference frames. In this general setting, it is important to notice that registering an event, which has been observed in the context of a reference frame, is not always equivalent to conferring a numerical identity to it by means of a real value corresponding to a physical attribute. On the contrary, the latter is only a *limited case* of the localization process, when, in particular, it is assumed that all reference contexts can be contracted to points. This is exactly the crucial assumption underlying the employment of the set-theoretic structure of the real line as a model of ‘physical continuum.’ The consequences of this common assumption have posed enormous problems for the interpretation of quantum theories, which are characterized by different localization schemes for the individuation of events. More concretely, set-theoretic axiomatizations of quantum event structures tend to hide the intrinsic significance of *Boolean* localizing contexts in the formation of these structures.

The operational procedures followed in quantum measurement are based explicitly on the employment of appropriate Boolean environments. The construction of these contexts of observation is related to our specific measurement arrangements and can be metaphorically considered as Boolean pattern recognition mechanisms. In this way, we may argue that the real significance of a quantum event structure proves to be, not at the level of events, but at the level of the specific form of interlocking or gluing together overlapping Boolean localizing contexts of observation into a globally coherent whole. From a logical perspective, the *gluing mechanism* accompanying quantum observable localization schemes is effectively a novel form of *logical internal relation*, which is absent classically, but plays a fundamental role in conceptualizing quantum-theoretic predication in topological terms. This is crucial for modern physics because topology is uniquely appropriate to the depiction of the transition from the *local* to the *global*, and vice versa, without using any metrical notions. For example, the significance of localization and gluing is of paramount importance for making sense of some unusual *topological quantum effects*, like quantal phase factors (the Berry phase) and the Aharonov-Bohm effect, as well as

for understanding *decoherence* in the framework of the consistent histories approach to quantum theory, via coarse-graining localization processes.¹

A natural question arising in this setting, inspired by the various interpretational problems of quantum theory, is the following: Is there an alternative conception of the ‘physical continuum’ that does not rely on an assumed preexisting set-theoretic structure of points on the real line? This question properly challenges the hypothesis that our form of observation is tautological with set-theoretic real number representability. In order to realize the necessity of an evolution in the semantics of the ‘physical continuum’ we emphasize that we have to consider carefully the different notions of localization associated with physical observables in classical and quantum measurement processes, respectively. The elucidation of this difference induces a natural evolution in the semantics of the physical continuum from a set-theoretic to a sheaf-theoretic one, as will be explained presently.

6.2 Localization Schemes

Before discussing the functioning of localization processes or schemes related with physical observables, it is necessary to include a brief intuitive description of ideas related with the notion of a physical continuum. For this purpose, the following quotations from Henri Poincaré² and Alfred North Whitehead,³ respectively, are particularly helpful:

We cannot say that our element is without extension, since we cannot distinguish it from neighboring elements and it is thus surrounded by a sort of haze. If the astronomical comparison may be allowed, our “elements” would be like nebulae, whereas the mathematical points would be like stars.

The ultimate facts of nature are events. . . The constitutive character of nature is expressed by the “contingency of appearance” and the “uniform significance of events”. . . By extension we mean that quality by virtue of which one event may be part of another, or two events may have a common part. Nature is a continuum of events so that any two events are both parts of some larger event.

The primary conception of a ‘physical continuum’ constitutes an inexhaustible complex of overlapping and non-overlapping events. Such an *event ontology* lies at the basis of our physical theories and necessitates a careful analysis of its premises before the acceptance of a particular mathematical model as a representational device for its functioning. The consideration of the notion of ‘event’ as a

primary concept in the comprehension of the ‘physical continuum’ poses immediately the following question: How are events to be *related* to each other? If the continuity of nature is to be ascribed to the relations among events, then the fundamental relation is *extension*. The relata in the relation of extension are the events, such that each event is part of a larger whole and each event encompasses smaller events. Extension is also inextricably tied with the assumption of *divisibility* of events signifying a *part-whole* or *local-global* type of relation. Thus, the crucial role in the conceptualization of event ontology is played by the topological notion of a localization scheme, inevitably endowed with an internal relational gluing structure, such that there exists a consistency in the *inverse processes* of *extending* or *inducing* observables from the local or partial to the global, and conversely, *restricting* or reducing observables from the global to the local. The internal relational gluing structure of a localization scheme expresses the fundamental idea that “the primary relationship of physical occasions is extensive connection.”⁴

It is important to emphasize, again, that only via a localization scheme does it become possible to discern events and assign an individuality to them. The formulation of the relations of extension and restriction via the physical and mathematical notion of a localization scheme elucidates the following excerpts from Whitehead’s *Process and Reality*, seen earlier in chapter 5:

The real potentialities relative to all standpoints are coordinated as diverse determinations of one extensive continuum. This extensive continuum is one relational complex in which all potential objectifications find their niche. It underlies the whole world, past, present, and future. Considered in its full generality, apart from the additional conditions proper only to the cosmic epoch of electrons, protons, molecules, and star-systems, the properties of this continuum are very few and do not include the relationships of metrical geometry. An extensive continuum is a complex of entities united by the various allied relationships of whole to part, and of overlapping so as to possess common parts, and of contact, and of other relationships derived from these primary relationships. This extensive continuum expresses the solidarity of all possible standpoints throughout the whole process of the world. It is not a fact prior to the world; it is the first determination of order—that is, of real potentiality—arising out of the general character of the world. . . All actual entities are related according to the determinations of this continuum; and all possible actual entities in the future must exemplify these determinations in their relations within an already actual world.⁵

Extension, apart from its spatialization and temporalization, is that general scheme of relationships providing the capacity that many objects can be welded into the real unity of

one experience. Thus, an act of experience has an objective scheme of extensive order by reason of the double fact that its own perspective standpoint has extensive content, and that the other actual entities are objectified with the retention of their extensive relationships. These extensive relationships are more fundamental than their more special spatial and temporal relationships. Extension is the most general scheme of real potentiality, providing the background for all other organic relations.⁶

The general purpose of a localization scheme amounts to filtering the information contained in a global structure of partially ordered physical events, through an appropriate *algebraic structure of localizing frames*, which can be determined by a *homologous* operational physical procedure of measurements. In this way, the substantial constituents of a localization scheme should specify the kind of loci of variation (or equivalently, reference contexts) to be used for observation of events. These contexts should instantiate generalized *reference frames* such that reference to concrete events of the specified kind can be made possible with respect to them. The requirements accompanying their instantiation are the following: First, they should constitute expressions of event ontology respecting the inverse relations of extension and restriction. Second, they should *not* be based on an underlying structure of set-theoretic points on the real line. This equivalently means that localization schemes used for the individuation of events should not depend on the existence of points. Third, due to their determination by a homologous operational physical procedure of measurements the *internal relations* of these varying reference contexts should play a crucial role in the coherent action of a localization scheme, and thus should not be neglected in its initial specification. According to Whitehead:

The extensive scheme is nothing else than the generic morphology of the internal relations which bind the actual occasions into a nexus, and which bind the prehensions of any one actual occasion into a unity, coordinately divisible.⁷

Interpreted mathematically, this essentially means that the reference contexts of a localization scheme should not only form a class, but they should form a *category*. More concretely, *the reference contexts together with their structural transformations should form a mathematical category*⁸ of localizing frames. For example, a localization scheme, which is based on the process of coarse-graining, should be implemented by a partial ordering relation among the localizing frames, which is obviously a category, where the only possible arrow between any two localizing frames is an inclusion. It is necessary to emphasize that the kind of reference contexts or loci of variation, together with their structural internal relations, determines precisely the concrete categorical environment employed

operationally by a homologous operational physical procedure of measurements. An instance of this is a category of open sets, ordered by inclusion, in a topological measurement space. Another instance, of particular significance in quantum theory, is a category of Boolean localizing algebraic contexts of quantum measurements, which may be considered as local Boolean subalgebras of a global quantum observable algebra.⁹

6.3 The Conceptual Framework of Category Theory

6.3.1 The Necessity of a Categorical Framework

Category theory provides a general theoretical framework for dealing with systems formalized through appropriate mathematical structures emphasizing their *mutual relations* and *transformations*. The central focus of the categorical way of rethinking basic notions can be described as a shift in the emphasis of what is considered to be fundamental for the formation of structures. In the set-theoretic mode of thinking, structures of any conceivable form are defined as sets of elements endowed with appropriate external relations. In the category-theoretic mode, the emphasis is placed on the transformations among the objects of a category devised to represent its internal relational structure by means of appropriate constraints on the collection of these transformations. In this sense, the notion of structure does not refer exclusively to a fixed universe of sets of predetermined elements, but acquires a *variable reference*.

The major foundational significance of category theory regarding our present enquiry on the problem of modeling the physical continuum, according to the requirements posed previously, is related to the following realization: The effect of casting a mathematical construct for modeling the physical continuum in category-theoretic language is equivalent to conferring a degree of ambiguity of reference on this construct. This is an appealing aspect for our purposes since it is exactly the doctrine of employing these constructs to depict a fixed, absolute universe of sets that we wish to over throw. According to category theory the mathematical constructs used for our modeling purposes should be regarded as possessing meaning only in relation to local frameworks, thus replacing the notion of unique reference to the universe of sets with other legitimate and varying frameworks of interpretation.¹⁰ This means, in turn, that a model of the physical continuum can be determined only in relation to a category and, most important, this category may vary.

We claim that it is exactly the referential ambiguity of the construct representing the physical continuum, interpreted in varying categorical environments, that is needed in order to accomplish the

critical evolution away from the conventionally assumed preexisting structure of points representing a true ‘physical continuum’ of events—an evolution that is of paramount importance to modern fundamental physics. Of course, the ambiguity of reference after casting the construct in categorical language should be fixed by its applicability in concrete categorical environments. Thus, the instantiation of these reference contexts should, from a physical viewpoint, depend on the *operational procedures* devised for the individuation of events in the continuum. In this sense, the notion of a localization scheme becomes a fundamental ingredient in modeling the physical continuum because it depicts the appropriate categorical environment suited to express its meaning, in agreement with the operational procedures followed by distinct physical theories.

Apart from determining the appropriate categorical environments for the operational modeling of the physical continuum, localization schemes referring directly to events likewise play the equally important role of representing a *base structure of varying loci* of the categorical kind depicted—a base structure over which the *partial order* of the extended continuum of events should *fiber*. Notice that in the set-theoretic real line model the role of such a locus is played only by a point, since a point conferring numerical identity to an event is the unique idealized localization measure. Consequently, from a topological perspective, adopting this assumption of an *atomic topology* on the continuum set is responsible for the conception of observable events as sharply distinguishable in classical theories. Indeed, from such a perspective, the fibering of an event continuum over a base structure of varying loci, representing concrete localization schemes in a topological context, can provide for such distinguishability and yet account for possible variations in the topology as well.

It is instructive to make clear that the process of casting a concept in categorical language is achieved by the characterization of the concept in terms of an *arrow-theoretic* formulation, and not by a specification in terms of a set equipped with certain operations and satisfying axioms expressible as relations among its elements. In this sense, the adoption of the categorical syntax involves at least two important conceptual shifts away from typical thinking about mathematical constructs considered for the modeling of the physical continuum. The first shift concerns the primary significance of *structure-preserving mappings*, referred to as arrows in category theoretic language. This shift reflects an opposition to the usual prevailing significance of sets with structure, referred to as objects, correspondingly. The second shift concerns the meaning of a *universal construction*, which is made precise in the categorical formalism, and moreover constitutes the main objective of the latter in the syntactical terms of this theory.

6.3.2 Categorical Principles and Terminology

The basic categorical principles that we adopt for this application are summarized as follows:

- [i] To each kind of mathematical structure used to represent systems, there corresponds a *category* whose objects have that structure, and whose morphisms preserve it.
- [ii] To any natural construction on structures of one kind, yielding structures of another kind, there corresponds a *functor* from the category of the first specified kind to the category of the second. The implementation of this principle is associated with the fact that a construction is not merely a function from objects of one kind to objects of another kind, but must preserve the essential relationships among objects.
- [iii] To each natural translation between two functors having identical domains and codomains there corresponds a *natural transformation*.
- [iv] To any canonical bidirectional functorial correlation between two kinds of mathematical structures there corresponds an *adjunction* (pair of *adjoint functors*) between the corresponding categories.

Categories:

A category C is a class of objects and morphisms of objects such that the following properties are satisfied:

- [1] For any objects X, Y all morphisms $f: X \rightarrow Y$ form a set denoted $Hom_C(X, Y)$;
- [2] For any object X an element $id_X \in Hom_C(X, X)$ is distinguished; it is called the identity morphism;
- [3] For arbitrary objects X, Y, Z , the set mapping is defined $Hom_C(X, Y) \times Hom_C(Y, Z) \rightarrow Hom_C(X, Z)$;
For morphisms $g \in Hom_C(X, Y)$, $h \in Hom_C(Y, Z)$ the image of the pair (g, h) is called the composition; it is denoted $h \# g$. The composition operation is associative;
- [4] For any $f \in Hom_C(X, Y)$ we have $id_Y \# f = f \# id_X = f$.

For an arbitrary category C the opposite category C^{op} is defined in the following way: the objects are the same, but $Hom_{C^{op}}(X, Y) = Hom_C(Y, X)$, that is, all arrows are inverted. A category C is called small if the classes of its objects and morphisms form genuine sets respectively.

Functors:

Let C, D be categories. A covariant functor $F: C \rightarrow D$ is a class mapping that transforms objects to objects and morphisms to morphisms preserving compositions and identity morphisms:

$$\mathbf{F}(id_X) = id_{\mathbf{F}(X)}; \mathbf{F}(g \circ f) = \mathbf{F}(g) \circ \mathbf{F}(f)$$

A contravariant functor $\hat{\mathbf{F}}: \mathbf{C} \rightarrow \mathbf{D}$ is, by definition, a covariant functor $\mathbf{F}: \mathbf{C}^{\text{op}} \rightarrow \mathbf{D}$.

Natural Transformations:

Let $\mathbf{C}, \mathbf{D}$ be categories, and let further $\mathbf{F}, \mathbf{G}$ be functors from the category $\mathbf{C}$ to the category $\mathbf{D}$. A natural transformation τ from $\mathbf{F}$ to $\mathbf{G}$ is a mapping that assigns to each object A in $\mathbf{C}$ a morphism τ_A from $\mathbf{F}(A)$ to $\mathbf{G}(A)$ in $\mathbf{D}$, such that for every arrow $f: A \rightarrow B$ in $\mathbf{C}$ the following diagram in $\mathbf{D}$ commutes:

$$\begin{array}{ccc} \mathbf{F}(A) & \xrightarrow{\tau_A} & \mathbf{G}(A) \\ \mathbf{F}(f) \downarrow & & \downarrow \mathbf{G}(f) \\ \mathbf{F}(B) & \xrightarrow{\tau_B} & \mathbf{G}(B) \end{array}$$

That is, for every arrow $f: A \rightarrow B$ in $\mathbf{C}$ we have:

$$\mathbf{G}(f) \circ \tau_A = \tau_B \circ \mathbf{F}(f)$$

Natural Isomorphisms:

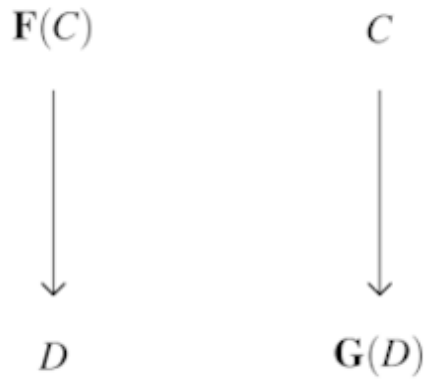
A natural transformation $\tau: \mathbf{F} \rightarrow \mathbf{G}$ is called a natural isomorphism if every component τ_A is invertible.

Adjoint Functors:

Let $\mathbf{F}: \mathbf{C} \rightarrow \mathbf{D}$ and $\mathbf{G}: \mathbf{D} \rightarrow \mathbf{C}$ be functors. We say that $\mathbf{F}$ is left adjoint to $\mathbf{G}$ (and correspondingly that $\mathbf{G}$ is right adjoint to $\mathbf{F}$), if there exists a bijective correspondence between the arrows $\mathbf{F}(C) \rightarrow D$ in $\mathbf{D}$ and $C \rightarrow \mathbf{G}(D)$ in $\mathbf{C}$, which is natural in both C and D .

$$\mathbf{F}: \mathbf{C} \in \mathbf{D}: \mathbf{G}$$

Pictorially we have:



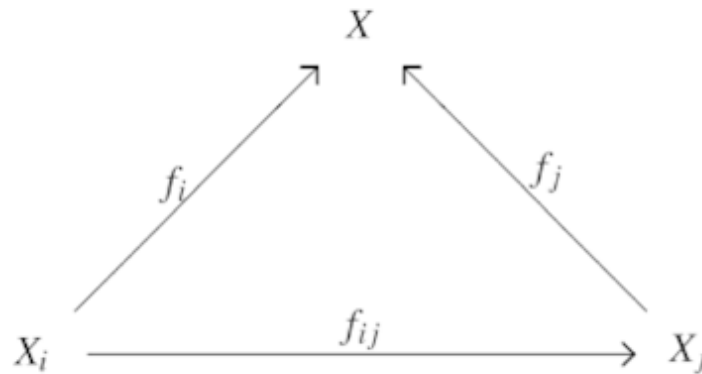
where the left part is in D and the right in C . Then, we say that the above pair of adjoint functors constitute a categorical adjunction.

Diagrams:

A diagram $X = (\{X_i\}_{i \in I}, \{F_{ij}\}_{i,j \in I})$ in a category C is defined as an indexed family of objects $\{X_i\}_{i \in I}$ and a family of morphism sets $\{F_{ij}\}_{i,j \in I} \subseteq \text{Home}_C(X_i, X_j)$.

Cocones:

A *cocone* of the diagram $X = (\{X_i\}_{i \in I}, \{F_{ij}\}_{i,j \in I})$ in a category C , consists of an object X in C , and for every $i \in I$, a morphism $f_i : X_i \rightarrow X$, such that $f_i = f_i \# f_{ij}$ for all $j \in I$, that is, such that for every $i, j \in I$, and for every $f_{ij} \in F_{ij}$ the diagram below commutes:



Colimits:

A colimit (*inductive limit*) of the diagram $X = (\{X_i\}_{i \in I}, \{F_{ij}\}_{i,j \in I})$ is a cocone with the property that for every other cocone given by morphisms $f_i : X_i \rightarrow X'$, there exists exactly one morphism $f : X \rightarrow X'$, such that $f_i = f \# f_i$, for all $i \in I$ (universality property).

Reversing the arrows in the above definitions of cocone and colimit of a diagram $X = (\{X_i\}_{i \in I}, \{F_{ij}\}_{i,j \in I})$ in a category C , results in the dual notions called *cone* and *limit (projective limit)* of X respectively. Moreover, starting with a diagram $X = (\{X_i\}_{i \in I}, \{F_{ij}\}_{i,j \in I})$ in a category C that consists only of the objects X_i , $i \in I$, as nodes but without morphisms, that is all $F_{ij} = \#$, we obtain the notion of the categorical *coproduct*, $\#_{i \in I} X_i$ (as a special colimit) and *product*, $\prod_{i \in I} X_i$ (as a special limit) respectively. The morphisms f_i in the corresponding definitions are called *canonical injections* of the coproduct and *canonical projections* of the product, respectively. We emphasize that we can derive special notions of limits and colimits, corresponding to the *shape* of the base diagram X . In this sense we obtain the following; an *initial object* is the colimit of the diagram consisting of the empty set. A *coequalizer* is the colimit of a diagram consisting of two parallel arrows $A \rightrightarrows B$. A *pushout* is the colimit of a diagram of the form:

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \downarrow g & & \\ & & C \end{array}$$

The dual notions are the following: a *terminal object* is the limit of the diagram consisting of the empty set. An *equalizer* is the limit of a diagram consisting of two parallel arrows $A \rightrightarrows B$. A *pullback* is the limit of a diagram of the form:

$$\begin{array}{ccc} & & B \\ & & \downarrow f \\ C & \xrightarrow{g} & A \end{array}$$

6.3.3 Universality and Equivalence

The definition of a category of algebraic objects of some kind with arrows being *homomorphisms* (structure-preserving morphisms) between them may be thought as an abstraction of the behavior of functions closed under the associative operation of composition. More precisely, the notion of a function is generalized to the notion of a homomorphism, whereas the associative operation of composition becomes an operation on sets of homomorphisms between algebraic objects of the same kind satisfying the same properties that functions and compositions satisfy. Notice that the composition of two functions f and g , denoted as $f \# g$, is defined only in the case that the codomain of g is the domain of f . Moreover, the composition of a function f with the identity of either its domain or codomain gives f again.

In a nutshell, the notion of a category of algebraic objects of some kind is a conception based on the behavior of *functions* closed under the associative operation of composition, abstracted in terms of homomorphisms, which in turn, have been idealized as algebraic ‘generalized elements’ (or their duals) determining completely the algebraic objects themselves. Because there exists an obvious *duality* between *incoming* and *outgoing* homomorphisms with respect to an algebraic object, this is built into the definition of a category so that the operation of *arrow reversal* leaves *invariant* the concept of a category, meaning that this operation gives again a category being in dual or opposite relation in comparison to the given one. A consequence of this fact is that all categorical constructions come in *dual pairs* corresponding to the dual viewpoints of considering incoming or outgoing arrows with respect to a constituted algebraic object.¹¹

The foundational significance of category theory for our purposes stems from the fact that it provides a precise mathematical framework to express the basic Whiteheadian structure of the relational realist philosophical scheme. In particular, the two most fundamental concepts epitomizing the philosophical function of the categorical framework in relation to our objectives are the concepts of *universality* and *equivalence*. More concretely, category theory provides a framework of identification of universals in mathematical terms—that is, the instances of a property which exemplify the property in such a paradigmatic way that all other instances have this property by virtue of factorizing uniquely through the corresponding universal. In this sense, objects are specified categorically via some universality property and only up to equivalence (canonical isomorphism).

The universal for a property represents the essential characteristic of a property in the abstractive interpretational sense of the Aristotelian *aphairein*. For example, the categorical notion of the inductive limit is a universal, viz., it is characterized by a universal mapping property, which depicts it *uniquely up to equivalence*. It is instructive to think about the categorical notion of inductive limit as a universal process which exemplifies algebraically Whitehead’s method of *extensive abstraction*. We note that the essence of the latter was its formulation of a mathematical procedure by which the *approximation to an*

ideal limit is consistently achieved depending on the properties of the relation of extension. The precise term coined by Whitehead in order to describe universals of this categorical nature is “eternal object” or “pure potential for the specific determination of fact.” An eternal object is always a *potentiality* for actual entities, and thus has a metaphysical status which is distinct from actuality, although it cannot exist separately from it. In particular, each eternal object / pure potential is characterized by its mode of *ingression* into some concrete actual occasion. Thus its functioning as a potentiality for actuality reveals its *relational essence*. Following Whitehead’s description:

A member of this species can only function relationally: by a necessity of its nature it is introducing one actuality, or nexus, into the real internal constitution of another actual entity. Its sole avocation is to be an agent in objectification. It can never be an element in the definiteness of a subjective form. The solidarity of the world rests upon the incurable objectivity of this species of eternal objects.^{[12](#)}

The crucial semantic convergence between the category-theoretic notion of a *universal* with Whitehead’s notion of an *eternal object* comes from the realization that both of them act as agents in the objectification of a multiplicity of actual occasions as a unified relational structure. For Whitehead, this is a categorical feature of relations among actual occasions, referred to as ‘transmutation.’^{[13](#)}

In category-theoretic terms the operation of the category of transmutation is subsumed by the action of pairs of adjoint functors forming adjunctions. This is the case because all *categorical universals* may be thought of as defined up to canonical isomorphism by the existence of pairs of *adjoint functors* between appropriate categories. In a well-defined sense adjoint functors should be understood as generalized *conceptual inverses* to each other. For example, the conceptual inverse of an ‘extension functor’ from one algebraic category to another is called a ‘restriction functor’ pointing in the inverse direction. Likewise, an adjunction between two categories gives rise to a family of *universal morphisms*, one for each object in the first category and one for each object in the second. In this way, each object in the first category induces a certain property in the second category and the universal morphism carries the object to the universal for that property; that is, it transmutes it to an agent of objectification in the second category exemplifying this property in a paradigmatic way. Every adjunction extends to an *adjoint equivalence* of certain subcategories of the initially correlated categories.

In general, the equivalence of two categories D and A is defined as follows: An *equivalence* of two categories D and A is defined by means of a pair of opposite directing functors; $F : D \rightarrow A$, $G : A \rightarrow D$ and a pair of *natural isomorphisms*; $\tau : 1_D \rightarrow G \# F$, $\rho : 1_A \rightarrow F \# G$ where τ is a *natural isomorphism* of

the identity functor on D , and p is a natural isomorphism of the identity functor on A . In this case, the functorial process $\mathbf{F}: D \rightarrow A$ is called *strictly inverse* to the functorial process $\mathbf{G}: A \rightarrow D$. If the pair of opposite directing functors is adjoint, then the equivalence is called an adjoint equivalence. The notion of equivalence of categories thus transcribes the general concept of *congruence* in a categorical /functorial context.

Now we make the observation that given any category A , the morphisms in A , or equivalently the morphisms in A^{op} , can be defined as the elements in the values of the $\mathbf{Hom}_A$ - bifunctor:

$$\mathbf{Hom}_A(-, -) = \mathbf{y}_{(B)}^{(A)} = A^{op} \times A \rightarrow \mathbf{Sets}$$

from the product category $A^{op} \times A$ to the category of sets. This is because A and A^{op} are considered to have the same objects and reversed morphisms. The $\mathbf{Hom}_A$ -bifunctor is such that:

- i. By fixing an object B in A^{op} , $\mathbf{y}_B: A \rightarrow \mathbf{Sets}$ is the *covariant representable functor*, represented by B , and defined as follows:

[1] For all objects X in A , $\mathbf{y}_B(X) := \mathbf{Hom}_A(B, X)$.

[2] For all homomorphisms $f: X \rightarrow Y$ in A ,

$$\mathbf{y}_B(f): \mathbf{Hom}_A(B, X) \rightarrow \mathbf{Hom}_A(B, Y)$$

is defined as pre-composition with f viz., $\mathbf{y}_B(f)(g) := f \# g$.

- ii. Correspondingly by fixing an object A in A , $\mathbf{y}^A: A^{op} \rightarrow \mathbf{Sets}$ is the *contravariant representable functor*, represented by A , and defined as follows:

[1] For all objects X in A^{op} , $\mathbf{y}^A(X) := \mathbf{Hom}_{A^{op}}(X, A)$.

[2] For all homomorphisms $f: Y \rightarrow X$ in A^{op} ,

$$\mathbf{y}^A(f): \mathbf{Hom}_{A^{op}}(X, A) \rightarrow \mathbf{Hom}_{A^{op}}(Y, A)$$

is defined as pre-composition with f viz., $\mathbf{y}^A(f)(g) := g \# f$.

The fundamental fact is that an object A in $\mathcal{A}$ is *entirely retrievable up to equivalence* (canonical isomorphism) from knowledge of the representable functor $\mathbf{y}^A : \mathcal{A}^{op} \rightarrow \mathbf{Sets}$ (or equivalently from $\mathbf{y}_A : \mathcal{A} \rightarrow \mathbf{Sets}$). This fact, a consequence of a result known as the *Yoneda Lemma*,¹⁴ can be expressed this way:

Let A, B be objects in a category $\mathcal{A}$. Suppose we are given an isomorphism of their associated functors: $\mathbf{y}^A \cong \mathbf{y}^B$. Then there is *a unique isomorphism of the objects themselves*, that is $A \cong B$, which gives rise to this isomorphism of functors. The conceptual significance of this result is that it has the following relational process-theoretic interpretation: An object A of a category $\mathcal{A}$ is *determined uniquely up to equivalence by the network of all internal relations* that the object A has with all the other objects in $\mathcal{A}$. Guerino Mazzola puts it in the following way:

The revolutionary idea of “understanding by changing perspective” was introduced. . . In its common sense rendering, it states that in order to understand an object, you just have to walk round the object. This seemingly trivial insight was put into a rigorous mathematical shape by Nobuo Yoneda [Yoneda 1954]. His famous Yoneda lemma states that a mathematical object can be classified up to isomorphisms by its so-called functor; the latter is precisely the system of all “views” or “perspectives” of the given object from all other objects of the same structure type (category). This statement even allows to construct objects via their functors.¹⁵

6.4 The Methodology of Uniform Fibrations:

Variable Set Presheaves

From the preceding it is clear that a localization scheme can be precisely conceived as a *functorial* generalization of the notion of *functional dependence*. In the trivial case, the only locus is a point serving as a unique idealized measure of localization, and moreover the only kind of reference frame is the one attached to a point. Thus, in mathematical terms, the localization process should be understood as *an action* of some category of local or partial reference contexts on a set-theoretic global structure of physical events. The latter is thus *partitioned into sorts* parameterized by the objects of the category of contexts. In this sense, the functioning of a localization scheme can be represented by means of a *fibred construct*, understood geometrically as a *variable set over the base category of local reference contexts*. Analogous to the case of the action of a group on a set of points, the *fibers* of this construct may be

thought of as the ‘*generalized orbits*’ of the action of the category of local contexts. The notion of functional dependence incorporated in this action forces the partially ordered structure of physical events to fiber over the base category of local reference contexts.

From a physical perspective, the meaningful representation of a global ordered structure of events as a fibred construct, expressed in terms of an action of some localization scheme as depicted above, should also incorporate the requirement of *uniformity*. By uniformity we mean that for any two events observed over the same domain of measurement, the structure of all reference frames that relate to the first *cannot be distinguished* in any possible way from the structure of frames relating to the second. According to this principle, all the localized events within any particular reference context of a localization scheme should be *uniformly equivalent* to each other. The compatibility of the localization process with the requirement of uniformity demands that the relation of (partial) order in a global set-theoretic universe of events is induced by appropriately *lifting* a structured family of arrows from the base category of local reference contexts to the fibers. In technical terms, the representation of a global ordered structure of events as a fibred construct via the action of some localization scheme incorporating the requirement of uniformity is modeled by the concept of a *presheaf*, and in particular by the category of its elements¹⁶ as it will be precisely formalized later in Part II.

The meaning of uniformity with respect to the above is depicted nicely by Nicholas Rescher in the following passage:

With concreta and existence in a world, this feature of uniformity and homogeneity must come to terms with differentiation and distinctiveness. Here every constituent substance must indeed look out upon exactly the same overall scene—exactly the same world—but it now does so from its own distinctively characteristic point of view. Like a line or plane, there is a complete uniformity and homogeneity, but it is not the uniformity of undifferentiated sameness but rather that of synoptic harmonization. And it is the feature of uniformity and homogeneity—of coordinating overall all of those different points of view—that is critical. For just this cosmic feature of reciprocal adjustment, of [Leibniz’s] *harmonie universelle* is one of the definitive factors for this world’s perfection.¹⁷

6.5 The Localization Role of Topology:

Covering Sieves and Sites

Having established the concept of a local reference context, defined according to the operational requirements of a corresponding homologous and uniform physical procedure of measurement, we can

now proceed to describe the mechanism by which observable information, defined with respect to overlapping local reference contexts, can be glued together by appropriate means. The gluing procedure is the topological manifestation of Whitehead's notion of extensive connection:

The "extensive" scheme is nothing else than the generic morphology of the internal relations which bind the actual occasions into a nexus, and which bind the prehensions of any one actual occasion into a unity, coordinately divisible. . . The primary relationship of physical occasions is extensive connection.^{[18](#)}

The topological character of the gluing procedure representing the process of extensive connection justifies its priority over the metrical notions of space and time. Likewise, for Whitehead:

In this general description of the states of extension, nothing has been said about physical time or physical space, or of the more general notion of creative advance. These are notions which presuppose the more general relationship of extension. They express additional facts about the actual occasions. The extensiveness of space is really the spatialization of extension; and the extensiveness of time is really the temporalization of extension.^{[19](#)}

Mathematically, the requirement of a consistent gluing procedure is implemented by the methodology of *completion* of a presheaf, or equivalently, its *sheafification*^{[20](#)} In this way, the structure of a sheaf arises by imposing upon the corresponding presheaf the following two requirements:[i] *Compatibility of observable information under restriction from the global to the local level*, and [ii] *Compatibility of observable information under extension from the local to the global level*. If we take together the requirements enforcing a representation of a global partially ordered structure of physical events as a uniform fibred construct over a base category of local reference contexts, localization schemes of the former are precisely modeled in the syntactical terms of *sheaves*. In this perspective, the evolution in the semantics of physical events from a set-theoretic description to a sheaf-theoretic one is completely justified, as will become clear throughout the remainder of part II. Moreover, the technical categorical notion of a topological sheaf captures mathematically Whitehead's conception of an extensive continuum as given in the passage quoted earlier:

The extensive continuum is a complex of entities united by the various allied relationships of whole to part, and of overlapping so as to possess common parts, and of contact, and of other relationships derived from these primary relationships. . . It is the first determination of order. . . The properties of this continuum are very few and do not include the relationships

of metrical geometry. . . These extensive relationships are more fundamental than their more special spatial and temporal relationships . . . Extension, apart from its spatialization and temporalization, is that general scheme of relationships providing the capacity that many objects can be welded into the real unity of one experience.²¹

Up to now, we have qualified reference contexts in the base category as ‘local’ in a rather informal way in order to convey the intuitive nature of this qualification. In order to further justify the latter and clarify the precise functionality of a localization scheme, it is necessary to define a proper notion of *topology on the base category* of reference contexts. Then, the meaning of local, as well as the meaning of the transition from the local level to the global level acquires a precise connotation.

The notion of a *categorical or Grothendieck topology*²² on a base category of reference contexts $\mathcal{B}$ is a categorical generalization of a system of set-theoretic *covers* on a topology T , where a cover for $U \in T$ is a set $\{U_i: U_i \in T, i \in I\}$ such that the union $\bigcup U_i = U$. The generalization is achieved by noting that the topology ordered by inclusion is a poset (partially ordered set) category and that any cover corresponds to a collection of inclusion arrows $U_i \rightarrow U$. The notion of a Grothendieck topology on a base category $\mathcal{B}$, consisting of reference contexts B , can thus be presented in terms of appropriate covering devices admitting a functorial interpretation.

The concept of a categorical topology requires, first of all, the abstraction of the constitutive properties of localization in appropriate categorical terms, and then, the effectuation of these properties for the definition of localization schemes. Thus, the concept of a categorical topology epitomizes the meaning of a localization scheme. Regarding these objectives, the requisite abstraction is being implemented by means of covering devices on the base category of reference contexts, called in categorical terminology *covering sieves*. The constitutive properties of localization being abstracted categorically in terms of sieves, qualified as *covering sieves* are the following: [i] The covering sieves should be *covariant* under pullback operations, viz., they should *bestable* under change of a base reference context; [ii] The covering sieves should be *transitive*.

The notion of covering sieves generalizes topologically Whitehead’s *covering relations* formulated in terms of *abstractive sets* as discussed in chapter 5. From a physical perspective, it is instructive to think of covering sieves as generalized topological measures of localization in the continuum. Philosophically, they express the *generic morphology of the mereotopological internal relations constituting the process-theoretic scheme of extensive connection in the continuum*. The operation

assigning to each reference context of the base category a collection of covering sieves satisfying the closure conditions, stated previously, gives rise to the notion of a Grothendieck topology on the base category of reference contexts.

In application of these notions to quantum theory, we think of the base category of reference contexts as a category of Boolean algebraic contexts related to typical quantum measurement situations.²³ It is a physical fact that any quantum measurement situation involves the preparation of a Boolean algebraic reference context (conceived locally in a proper way) with respect to which a measurement value can be obtained operationally by the use of some measurement device. In this setting a fundamental admonition is given in the Kochen-Specker theorem,²⁴ according to which the complete description of a quantum mechanical system is impossible to produce by way of a single system of Boolean devices extended globally. By contrast, in every concrete quantum measurement context, the set of logical events that can be actualized in *that* context forms a Boolean algebra. This fact motivates the assertion that *a Boolean algebra within the partially ordered set of quantum logical events plays the role of a local logical reference frame relative to which a measurement result is coordinatized*. Thus, in the words of Arthur Eddington:

In Einstein's theory of relativity the observer is a man who sets out in quest of truth armed with a measuring-rod [a metrical concept]. In quantum theory he sets out with a sieve [a topological concept].²⁵

The construction of a suitable Grothendieck topology on the base category of contexts is significant for the following reasons: First, it explicitly elucidates, with great precision, the conceptions of local and global in a categorical measurement environment, such that these conceptions become *detached from their restricted spatial connotation in terms of geometric point-spaces*—the latter being wholly absent in the case of quantum theory. Instead, these conceptions are depicted exclusively in terms of *relational observable information*. Secondly, it permits the *gluing of local observable information terms into global ones* by utilization of the notion of a sheaf for a suitable Grothendieck topology.

This is most easily demonstrated by first exploring the general notion of sieves, introduced earlier, and then proceeding to focus on the notion of covering sieves. In preface, it is instructive to note at this stage that in the usual definition of a localization scheme induced by a topological space X we use the *open neighborhoods* U of a point in X ; such neighborhoods are actually inclusions $U \# X$. The neighborhoods U in topological spaces can be replaced by morphisms of reference contexts $C \rightarrow B$, *not* necessarily inclusions, and this can be done in any complete category. In effect, a covering of a topological space by open sets can be replaced by a new notion of covering, provided by a family of

morphisms targeting some object in a categorical environment, and satisfying the above conditions [i] and [ii], which abstract the decisive features of a localization process.

A *B-sieve* with respect to a reference context B in the base category $\mathcal{B}$ is a family S of $\mathcal{B}$ -morphisms targeting B , such that if $C \rightarrow B$ belongs to S and $D \rightarrow C$ is any $\mathcal{B}$ -morphism, then the composite $D \rightarrow C \rightarrow B$ belongs to S . We may think algebraically of a *B-sieve* as a right *B-ideal*. We notice that in the case of $\mathcal{O}(X)$, since $\mathcal{O}(X)$ -morphisms are inclusions of open loci, a right *U-ideal* is identical with a *downwards closed U-subset*.

We notice that if S is a *B-sieve* and $h : C \rightarrow B$ is any arrow to the reference context B , then: $h^*(S) = \{ f / \text{cod}(f) = C, (h \# f) \in S \}$ is a *C-sieve*, referred to as the *pullback* of S along h , or the *restriction of S* along h where $\text{cod}(f)$ denotes the codomain of f . We notice that for a context B in $\mathcal{B}$, the set of all arrows into B is a *B-sieve*, called the *maximal sieve* on B , and denoted by, $t(B) := t_B$.

As we proceed to the next stage of development, the key conceptual issue to be settled is the following: How is it possible to *restrict* or *reduce* the set of all *B-sieves* for each reference context B in $\mathcal{B}$, denoted by $\mathbf{K}(B)$, such that each *B-sieve* of the restricted set can acquire the interpretation of a *covering B-sieve* with respect to a generalized covering system? Equivalently stated, we wish to impose the satisfaction of appropriate conditions on the set of *B-sieves* for each context B in $\mathcal{B}$, such that the subset of *B-sieves* obtained, denoted by $\mathbf{K}_\chi(B)$, implements the constitutive properties of a localization process. In this sense, the *B-sieves* of $\mathbf{K}_\chi(B)$ for each locus B in $\mathcal{B}$, to be thought as *covering B-sieves*, can be legitimately used for the implementation of localization processes. The appropriate conditions depicting the *covering B-sieves* from the set of all *B-sieves*, for each reference context B in $\mathcal{B}$, are the following:

[1] We interpret an arrow $C \rightarrow B$, where C, B are contexts in $\mathcal{B}$, as a *figure* of B , and thus we interpret B as an *extension* of C in $\mathcal{B}$. It is a natural requirement that the set of all figures of B should belong in $\mathbf{K}_\chi(B)$ for each context B in $\mathcal{B}$.

[2] The *covering sieves* should be *stable* under pullback operations. This requirement means, in particular, that the intersection of *covering sieves* should also be a *covering sieve*, for each reference context B , in the base category $\mathcal{B}$.

[3] Finally, it would be desirable to impose: [i] a *transitivity* requirement on the specification of the *covering sieves*, such that, intuitively stated, *covering sieves* of figures of a context in *covering sieves* of this context, should also be *covering sieves* of the context themselves, and [ii] a requirement of *common refinement* of *covering sieves*.

If we take into account the above requirements we can define a *generalized covering system*, called a *Grothendieck topology*, in the environment of the base category $\mathcal{B}$ as follows:

A *Grothendieck topology* on B is an operation $\mathbf{J}$, which assigns to each reference context B in B , a collection $\mathbf{J}(B)$ of B -sieves, called covering B -sieves, such that the following three conditions are satisfied:

- [1] For every reference context B in B , the maximal B -sieve $\{g : \text{cod}(g) = B\}$ belongs to $\mathbf{J}(B)$ (*maximality condition*).
- [2] If S belongs to $\mathbf{J}(B)$ and $h: C \rightarrow B$ is a figure of B , then $h^*(S) = \{f: C \rightarrow B, (h \# f) \in S\}$ belongs to $\mathbf{J}(C)$ (*stability condition*).
- [3] If S belongs to $\mathbf{J}(B)$, and if for each figure $h: C_h \rightarrow B$ in S , there is a sieve R_h belonging to $\mathbf{J}(C_h)$, then the set of all composites $h \# g$, with $h \in S$, and $g \in R_h$, belongs to $\mathbf{J}(B)$ (*transitivity condition*).

As a consequence of the definition and conditions above, we can easily verify that any two B -covering sieves have a common refinement, that is: if S, R belong to $\mathbf{J}(B)$, then $S \cap R$ belongs to $\mathbf{J}(B)$.

As a first application we may consider the partially ordered set of open subsets of a topological space X , viewed as the *base category of open reference contexts*, $O(X)$. Then we specify that S is a covering U -sieve if and only if U is contained in the union of open sets in S . The above specification fulfills the requirements of covering sieves posed above, and consequently, defines a Grothendieck topology on $O(X)$.

A category of reference contexts B together with a Grothendieck topology $\mathbf{J}$, is called a *site*, denoted by $(B, \mathbf{J})$. The notion of a site generalizes the notion of a topological measurement state space in categorical terms and constitutes the *relational variable topological background* for the effectuation of a localization scheme for *filtering or coarse-graining or grading* the information contained in a global structure of partially ordered physical events. This filtering occurs via localizing structural frames determined by a homologous operational physical procedure of measurement.

As will be demonstrated throughout the remainder of Part II, this conceptual framework is uniquely suited to the coherent and consistent description of localization processes in quantum theory. The basic idea is to associate the functioning of a localization scheme in a global quantum event continuum with covering sieves of local Boolean contexts of quantum measurement. Before applying the machinery of sites to quantum theory, however, it is instructive to discuss in detail the important, simplest case of a localization scheme induced by a topology on a space.

Notes

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13. Whitehead, *Process and Reality*, 292.

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19. Whitehead, *Process and Reality*, 289.

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